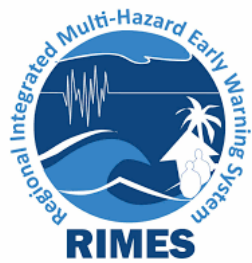




IMD RAINfall INformation System (IRAINS)



IRAINS APPLICATION GUIDE

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PREPARED FOR

IMD Hydromet

PREPARED BY

RIMES

INTRODUCTION

IRAINS (IMD RAInfall INformation System) is a specialized platform developed for the Hydromet Division, by IRU (IMD RIMES UNIT) using cutting edge technologies by focusing on the efficient tracking and analysis of rainfall data across various regions in India. This application serves as a central hub for rainfall data collected from multiple meteorological stations and is intended to support operations within the Hydromet Division. IRAINS offers a wide range of features that empower HQ, MC, and RMC users to monitor daily rainfall, examine historical data, and generate graphical and tabular reports, fostering transparency and accuracy in meteorological data handling. The system supports real-time data collection, statistical analysis, and customizable data entry, designed to meet IMD's mission of providing precise, accessible, and actionable climate information to its users. IRAINS is built with scalability in mind, handling data at multiple levels, from district and state to national coverage. It integrates interactive maps, extensive filtering options, and dissemination capabilities, enhancing data visibility and fostering timely responses to weather-related events. With an intuitive interface and comprehensive data analysis features, IRAINS is instrumental in addressing critical meteorological needs.

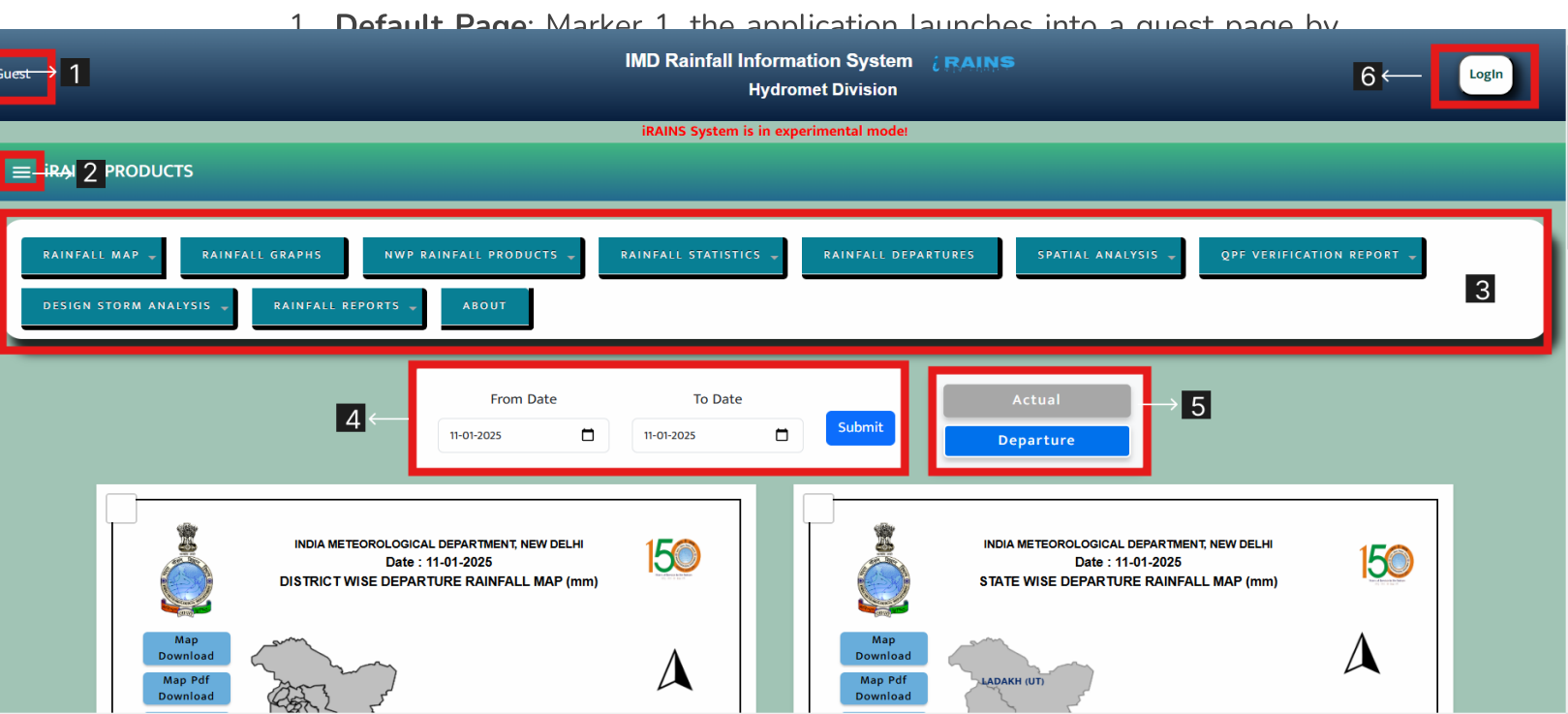
OVERVIEW:

The IRAINS Application Guide is a comprehensive document designed to help users understand and navigate the IRAINS platform effectively. IRAINS, developed by the IMD RIMES Unit, is a powerful tool for tracking and analyzing rainfall data across India. This guide explains the platform's key features, providing step-by-step instructions for users to access, filter, and analyze data efficiently.

The document highlights the importance of features like real-time data entry, verification, yearly statistics, significant rainfall analysis, and customizable reports. Each feature is explained with detailed markers, making it easier to locate and utilize them in the application.

By using this guide, users can enhance their understanding of the IRAINS platform, improving data accuracy and fostering timely decision-making. Whether for monitoring rainfall or generating detailed reports, this guide ensures users can leverage IRAINS to meet their meteorological needs effectively.

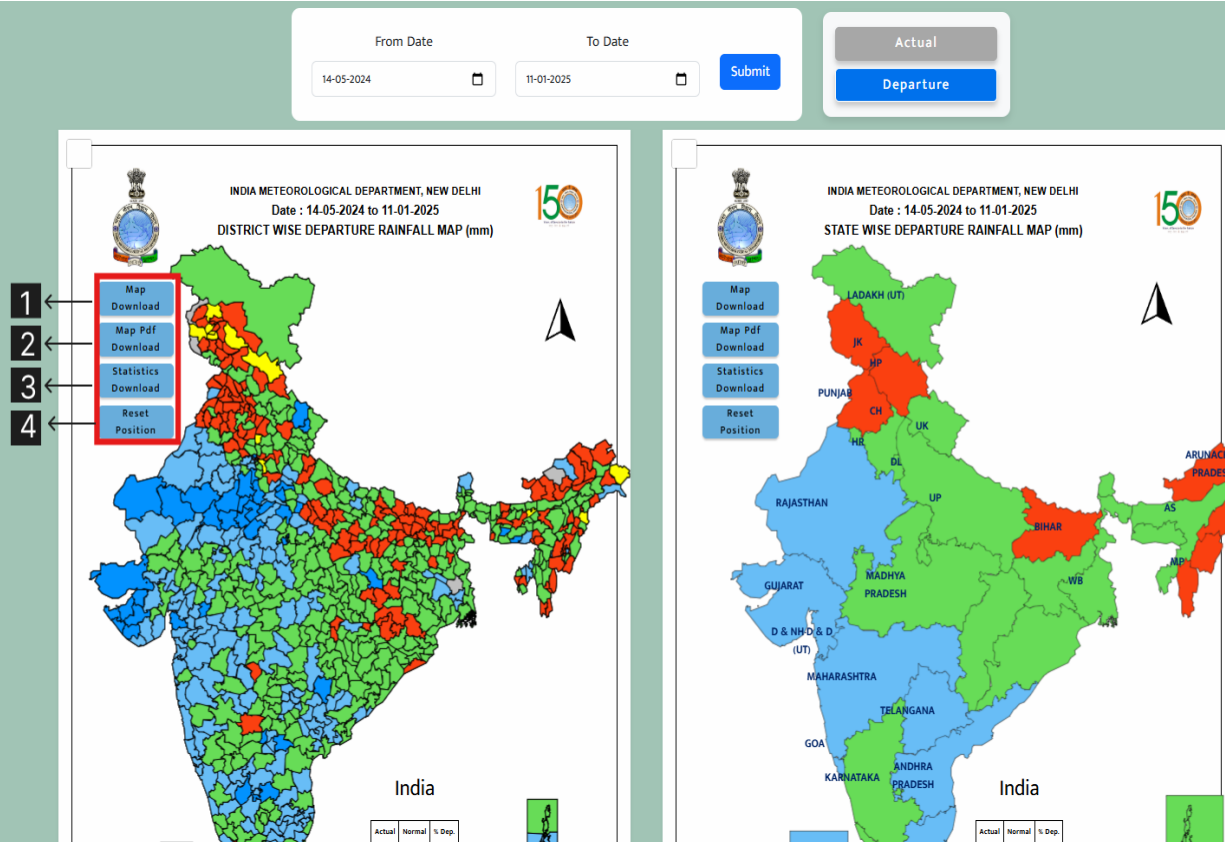
Figure 1 represents the default interface of the application when accessed as a guest user. The following features are highlighted and marked for clarity:



2. **View All Features Button:** Marker 2 highlights a button that allows guests to view all the features available to them.

3. **Guest Features List:** Marker 3 shows the navbar, list of features accessible to guest users.
4. **Date Selection Form:** Marker 4 identifies a form where users can select a "From Date" and "To Date" to filter and view desired results upon submission.
5. **Data View Options:** Marker 5 highlights options to toggle between viewing actual data or departure data.
6. **Login Button:** Finally, marker 6 highlights the login button, which enables users to access the application as authenticated users.

Figure 2 illustrates the various features and functionalities integrated into the map section of the application. The following features are highlighted:



1. **Map Download (PNG):** Marker 1 indicates the button to download map information in PNG format.
2. **Map Download (PDF):** Marker 2 points to the button for downloading map information in PDF format.
3. **Statistics Download:** Marker 3 shows the button to download statistical information related to the map.
4. **Reset Map View:** Marker 4 identifies the button used to reset the map view to its default position.

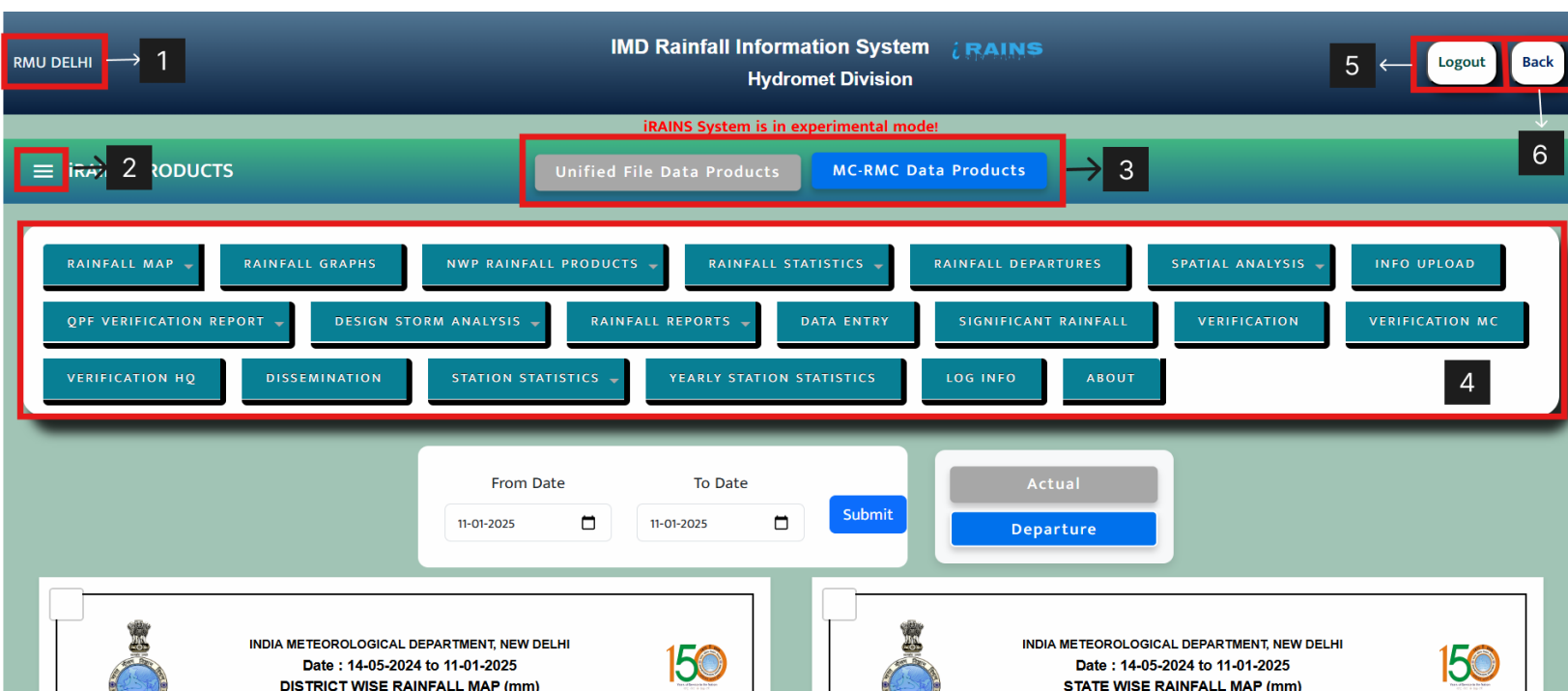
Figure 3 depicts the login page of the application, showcasing the following key features:

The screenshot shows the login interface of the iRAINS application. On the left, a banner features the iRAINS logo and the text 'India Meteorological Department RAINfall INformation System' over a background of water droplets. On the right, the header includes the IMD logo, the 150th anniversary emblem, and the text 'IMD RAINfall INformation System (iRAINS) Hydromet Division, India Meteorological Department, Ministry Of Earth Sciences, New Delhi-110 003'. The login form consists of an 'Email Address*' field (marked 1), a 'Password*' field (marked 2), a 'Remember Password' checkbox, a 'Forgot Password?' link, and a purple 'Login' button (marked 3). At the bottom, there are links for 'No account? Login as Guest' and 'To Register your account! Request here'.

1. **Email Address Field:** Marker 1, this input field allows users to enter their registered email address for authentication.

2. **Password Field:** Marker 2, this field is where users input their password to securely log in.
3. **Login Button:** Marker 3, this button submits the entered email and password for authentication. Upon clicking, users are logged into the application if the credentials are correct.

Figure 4: Interface After Successful User Login



1. **User Details :** Marker 1, Displays the details of the currently logged-in user accessing the system.
2. **Features Button :** Marker 2, On clicking this, users can view all the features available to them.
3. **Data Product Selection :** Marker 3, Provides two options: Unified Data Product and MC-RMC Data Product. Unified File Data Products compile MK rainfall data as a temporary solution until direct data entry is fully implemented by MC and RMC centers. This choice determines the format and scope of rainfall data displayed.

4. **Navbar** : Marker 4, Contains all the features accessible to the logged-in user, such as:

- 4.1. Rainfall Map
- 4.2. Rainfall Graph
- 4.3. NWP Rainfall Products
- 4.4. Rainfall Statistics
- 4.5. Rainfall Departures
- 4.6. Spatial Analysis
- 4.7. Info Upload
- 4.8. QPF Verification Report
- 4.9. Design Storm Analysis
- 4.10. Rainfall Reports
- 4.11. Data Entry
- 4.12. Significant Rainfall
- 4.13. Verification HQ
- 4.14. Dissemination
- 4.15. Station Statistics
- 4.16. Yearly Station Statistics
- 4.17. Log Info
- 4.18. About

5. **Logout Button** : Marker 5, Logs the user out of their profile and ends the session.

6. **Back Button** : Marker 6, Allows navigation back to the previous page or the home page when exploring features.

Data Entry : The Data Entry Security Feature enhances accountability and ensures secure access to the data panel. By requiring users to provide their officer ID or name, the system tracks and identifies the individuals making

Choose file

No file chosen

Upload Stations

[Sample File](#)

Choose file

INFALL	TYPE		LAT° N Eg : 21.3463	LNG° E Eg : 26.3453
Enter Officer ID/Name Officer Id/Name: Submit				

changes, such as adding or editing records.

The Data Entry interface allows users to manually input data before 12:30 PM. After this time, an automated email is sent to the respective MC and RMC centers, listing the station data that has not been updated. The following fields

Select Regions **1** ▼

Select MCs **2** ▼

Select RMCs **3** ▼

Select States **4** ▼

Select Districts **5** ▼

dd-mm-yyyy **6**

Submit **7**

STATIONS DATA **8**

Download **9**

Choose File No file chosen **10**

Upload Stations **11**

Choose File No file chosen **12**

Upload Rainfall **13**

S.NO	STATION ID	STATION NAME	RAINFALL	TYPE	NEW/OLD	Location Type	LAT° N Eg : 21.3463	LNG° E Eg : 26.3453	YEAR OF ACTIVATION	ACTION

Add Station **14**

are marked to get required data.

Select Region (Marker 1): Allows users to select the region for data entry.

Select MCs (Marker 2): Enables users to choose the MC (Meteorological Center).

Select RMCs (Marker 3): Used to select the RMC (Regional Meteorological Center).

Select States (Marker 4): Allows selection of the state for data entry.

Select District (Marker 5): Facilitates the selection of the district within the chosen state.

Select Date (Marker 6): Used to choose the date for which data is being entered.

Submit Button (Marker 7) : A button used to submit the selected data.

Download Station Data (Marker 8): A button to download station data in the required format.

Upload Station Detail File (Marker 9): Provides an option to upload a station details file in Excel format.

Upload Station Button (Marker 10): Used to upload the file selected in Marker 8 for station data.

Upload Rainfall Data (Marker 11): Allows users to upload rainfall data.

Upload Rainfall Button (Marker 12): Submits the file selected in Marker 10 for rainfall data upload.

Rainfall (Marker 13) : In data entry , the rainfall is a daily entry which we need to update daily.

Marker 14 : In marker 14, we have Type, NEW/OLD, Location type, Lat, Lng, Year of Activation., these fields are one time entry in the data panel.

Action Button(Marker 15) : In the action field , we have edit, delete option, where we can update the data or even can delete the data.

Add Station Button (Marker 12): Facilitates adding new station data by uploading a station file.

This section is crucial for ensuring timely updates of meteorological data and automating communication for missing entries.

Upload Station : It is a field in the data entry used to upload station data.

A	B	C	D	F	F	G	H	I
station_name	station_id	station_type	is_new_station	latitude	longitude	activationdate	centre_type	centre_nmae
GUNДАРDEHI	9010100000001	org	yes	31.652	80.2315	2024-02-21	mc	names
GURUR	9010100000002	aws	yes	32.652	81.2315	2024-02-22	mc	names
BALODA BAZAR	9010100000003	aws	yes	33.652	82.2315	2024-02-23	rmc	names
BHATAPARA	9010100000004	aws	yes	34.652	83.2315	2024-02-24	rmc	names
BILAI GARH	9010100000005	aws	yes	35.652	84.2315	2024-02-25	rmc	names
KASHDOL	9010100000006	aws	yes	36.652	85.2315	2024-02-26	rmc	names
PALLARI/PALARI	9010100000007	aws	yes	37.652	86.2315	2024-02-27	rmc	names
SIMGA	9010100000008	aws	yes	38.652	87.2315	2024-02-28	rmc	names
BALRAMPUR	9010100000009	aws	yes	39.652	88.2315	2024-02-29	rmc	names
BALRAMPUR	9010100000010	aws	no	40.652	89.2315	2024-03-01	rmc	names

The provided Excel format serves as a template for uploading station data into the system. It includes critical attributes necessary for defining and categorizing station details. The attributes cover a variety of station-specific information such as its unique identifier, geographical location, operational status, and associated center details. This format ensures structured and uniform data entry for efficient processing and management.

Attributes in Detail:

station_name: The name of the station.

station_id: A unique identifier for the station.

station_type: Specifies the type or category of the station.

is_new_station: A boolean field indicating whether the station is new (e.g., Yes/No or True/False).

Latitude: The geographical latitude coordinate of the station.

Longitude: The geographical longitude coordinate of the station.

activationofdate: The date when the station was activated or became operational.

centre_type: Specifies the type of center associated with the station (e.g., Regional, National).

centre_name: The name of the center associated with the station.

Upload Rainfall: It is a field in the data entry used to upload rainfall data.

[illegible]

Rainfall data entry in IRAINS is managed with specific guidelines and timelines to ensure accurate and timely updates. Data is uploaded using an Excel file containing attributes such as station_name, center_type, station_id, and dates with corresponding rainfall data. The privileges and responsibilities differ for HQ, RMC, and MC users. RMC and MC users can update data for up to 30 days, while HQ users have an extended privilege of 90 days to make updates. All data entry must be completed before 12:30 PM daily. If any RMC or MC fails to update the data by 12:30 PM, the system automatically sends an email reminder to the respective centers.

After the initial data entry, the verification window is open until 1:15 PM. Any data changes made after 1:30 PM will not be considered for the same day's map generation; instead, these updates will be reflected the following day. This ensures that finalized data is used for generating rainfall maps, maintaining the accuracy and integrity of the system.

Key Points:

Excel Format Attributes:

station_name, center_type, station_id, and dates.

Privileges:

RMC/MC: Can update data up to 30 days.

HQ: Can update data up to 90 days.

Timeline for Data Updates:

Data entry deadline: 12:30 PM.

Automatic email reminders: Sent after 12:30 PM for missing updates.

Verification window: 12:30 PM to 1:15 PM.

Changes after 1:30 PM are considered for the next day's map generation.

Data Changes:

Data finalized after 1:30 PM will not impact the current day's rainfall map.

These rules ensure structured workflows and timely communication within the rainfall data management system.

Significant Rainfall : The Significant Rainfall feature in the data panel is designed to efficiently filter, view, and export rainfall data based on specific criteria. Users can interact with the feature through several markers, each serving a distinct function to streamline data management.

RMU DELHI

IMD Rainfall Information System
Hydromet Division

RAINS

LogoutBack

1Select Rainfall Filter TypeCustom RangeNo Range

Select Date19-12-2024

Enter Rainfall Range0mm

3Submit

2

Significant Rainfall DataDownload4

Met Subdivision	District Name	Station Name	Rainfall (mm)
GUJARAT REGION	ARAVALLI	BAYAD	3.0
GUJARAT REGION	ARAVALLI	BAYAD_ARG	5.0
GUJARAT REGION	ARAVALLI	BHILODA	5.0
GUJARAT REGION	ARAVALLI	DHANSURA	6.0
GUJARAT REGION	ARAVALLI	MALPUR	6.0
GUJARAT REGION	ARAVALLI	MODASA	1.0

1. Select Rainfall Filter Type (Marker 1):

Provides the option to choose between two filter types:

- 1. **Custom Range:** Allows specifying a custom range for filtering rainfall data.
- 2. **No Range:** Displays all available data without applying range-based filters.

2. Search Filter:

Includes two input fields for specifying filter criteria:

- 1. **Select Date:** Allows the user to choose a specific date to filter data.
- 2. **Enter Rainfall Range:** Enables input of a range (e.g., minimum and maximum rainfall) for precise filtering.

3. Submit Button (Marker 3):

Filters the data based on the selected values in the **Rainfall Filter Type** and **Search Filter** fields.

4. Download Button (Marker 4):

Exports the filtered data into a PDF format, making it convenient for offline use or reporting.

This feature ensures users can effectively analyze rainfall data by customizing filters and exporting the required information seamlessly.

Verification HQ : The Verification HQ feature in the data panel provides an overview of station data, allowing users to analyze the status of stations based on verification and data update statuses. It offers a structured interface to filter, view, and assess station details efficiently.

IMD Rainfall Information System (iRAINS) Hydromet Division

[Back](#)

26-12-2024



1

2

[Submit](#)

STATIONS DATA

3

S.NO	MC or RMC	TOTAL STATIONS	UPDATES		VERIFICATIONS	
			UPDATED STATIONS	NOT UPDATED STATIONS	VERIFIED STATIONS	NOT VERIFIED STATIONS
1	MC RAIPUR	298	0	149	0	0
2	null null	888	0	444	0	0
3	MC BHOPAL	806	0	403	0	0
4	MC AHMEDABAD	732	0	366	0	0
5	RMC MUMBAI	576	0	288	0	0
6	MC BHUBANESWAR	634	0	317	0	0
7	RMC NAGPUR	300	0	150	0	0

1. Select Date (Marker 1): Allows users to choose a specific date to filter the station data.

2. Submit Button (Marker 2): Fetches the station data corresponding to the selected date upon clicking.
3. Station Data View (Marker 3): Displays the station data in a tabular format with the following fields:

S.No: Serial number for identification.

MC or RMC: Associated Meteorological Center (MC) or Regional Meteorological Center (RMC).

Total_Stations: Total number of stations within the selected criteria.

Updated Stations: Number of stations with updated data.

Not Updated Stations: Number of stations without updated data.

Verified Stations: Number of stations that are verified.

Not Verified Stations: Number of stations that are not verified.

This feature provides a comprehensive view of station data for monitoring and assessment, ensuring accurate reporting and efficient data management.

Yearly Station Statistics : The Yearly Station Statistics feature in the data panel provides detailed station statistics based on user-defined filters. It allows users to refine their search by selecting specific geographic and temporal criteria, enabling a focused analysis of station data.

RMU DELHI

IMD Rainfall Information System **RAINS**
Hydromet Division

Logout Back

Select Regions ▼ Select MCs ▼ Select RMCs ▼ Select States ▼ Select Districts ▼

1 2 3 4 5

From Date To Date

dd-mm-yyyy dd-mm-yyyy 6

Submit 7

Download 8

Sample selected data

CENTRAL INDIA

BHOPAL

Select RMCs

MADHYA PRADESH

ANUPPUR

From Date

To Date

26-12-2024

01-01-2025

Submit

Download

station_name	2024-12-25	2024-12-26	2024-12-27	2024-12-28	2024-12-29	2024-12-30
AMARKANTAK	0	0	0	16	0	0
ANUPPUR-AWS	0	0	0	5.3	0	0
BENIBARI	0	0	0	7.2	0	0
BIJURI	0	0	0	4	0	0
JAITHARI	0	0	0	0	0	0
KOTMA	0	0	0	0	0	0
PUSHPRAJGARH	0	0	0	6.4	0	0
VENKATNAGAR	0	0	0	3.3	0	0

- Select Region (Marker 1):** Enables the selection of a specific region for filtering station data.
- Select MCs (Marker 2):** Allows users to choose specific Meteorological Centers (MCs) for the search.
- Select RMCs (Marker 3):** Facilitates the selection of Regional Meteorological Centers (RMCs).
- Select State (Marker 4):** Provides the option to select a specific state for narrowing down station statistics.
- Select District (Marker 5):** Allows users to filter data based on a selected district.
- Filter Box (Marker 6): Date Filter Box:** Includes fields to specify a date range by selecting a **From Date** and **To Date**.
- Submit Button (Marker 7):** Filters and displays the station data based on the selected criteria upon clicking.
- Download Button (Marker 8):** Exports the filtered data into a PDF format for offline use and reporting.

This feature streamlines the analysis of yearly station statistics by offering flexible filters and export options for efficient data management.

SUPPORT CONTACTS:

If you have any doubts or need assistance using the application, please contact:

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